Rui CAI

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Education

The Chinese University of Hong Kong, Shenzhen, China Sep. 2023 – Present

B.E. in Computer Engineering, Expected May 2027 CGPA: 3.75/4.00 · Dean's List (2024–2025)

Selected Coursework: Machine Learning, Introduction to Robotics, Database Management Systems, Optimization, Data Structures and Algorithms, Computer Architecture

The Hong Kong University of Science and Technology, Hong Kong SAR Feb. 2026 – May 2026

Exchange Student

Selected Coursework: Computer Networks, Big Data Mining and Management, Learning, Reasoning and Decision Making in AI

Research & Project Experience

BELIEF-VLA: Long-Horizon Vision-Language-Action Learning, CUHK-Shenzhen Aug. 2026 – Present

Undergraduate Researcher Advisor: Prof. Guiliang Liu

- Developed a LaRA-VLA-inspired annotation pipeline for LIBERO demonstrations to generate structured subtask supervision for long-horizon robot manipulation
- Integrated the SimVLA model architecture into the BELIEF-VLA framework and adapted data-model interfaces for subsequent VLA experiments
- Participated in architecture design and experimental planning for block-autoregressive VLA models targeting long-horizon robot control

Independent Study in 3D/4D Computer Vision, CUHK-Shenzhen Sep. 2025 – Dec. 2025

Independent Study Advisor: Prof. Xiaoguang Han, GAP Lab

- Conducted an independent study in computer vision through paper reading, technical reports, and reproduction of published experiments, with a focus on 3D/4D Gaussian Splatting
- Implemented TaoGS-inspired modules within the DualGS framework and adapted the extended pipeline to an industry-provided proprietary dataset for data preprocessing and benchmark evaluation

Hull Tactical Market Prediction, Kaggle Competition Oct. 2025 – Dec. 2025

Machine Learning Engineer Team Project

- Built a reproducible time-series forecasting pipeline with a reusable FeatureStore for consistent preprocessing; trained and compared PatchTST and N-HiTS using MSE, IC, and Rank-IC
- Performed validation-driven model and ensemble selection and contributed to a GRPO-based policy for risk-aware position sizing
- Implemented an offline inference pipeline with persistent model loading, sliding-window caching, and cold-start handling under Kaggle runtime constraints

Skills

- **Programming:** Python, C++, SQL, MATLAB
- **ML & AI:** PyTorch, LightGBM, CatBoost, NeuralForecast; Reinforcement Learning, Time-Series Forecasting
- **Embodied AI & Vision:** Vision-Language-Action (VLA) Models, Robot Learning, 3D/4D Gaussian Splatting
- **Tools & Systems:** Linux, Git, Docker, Bash, MySQL, SQLite